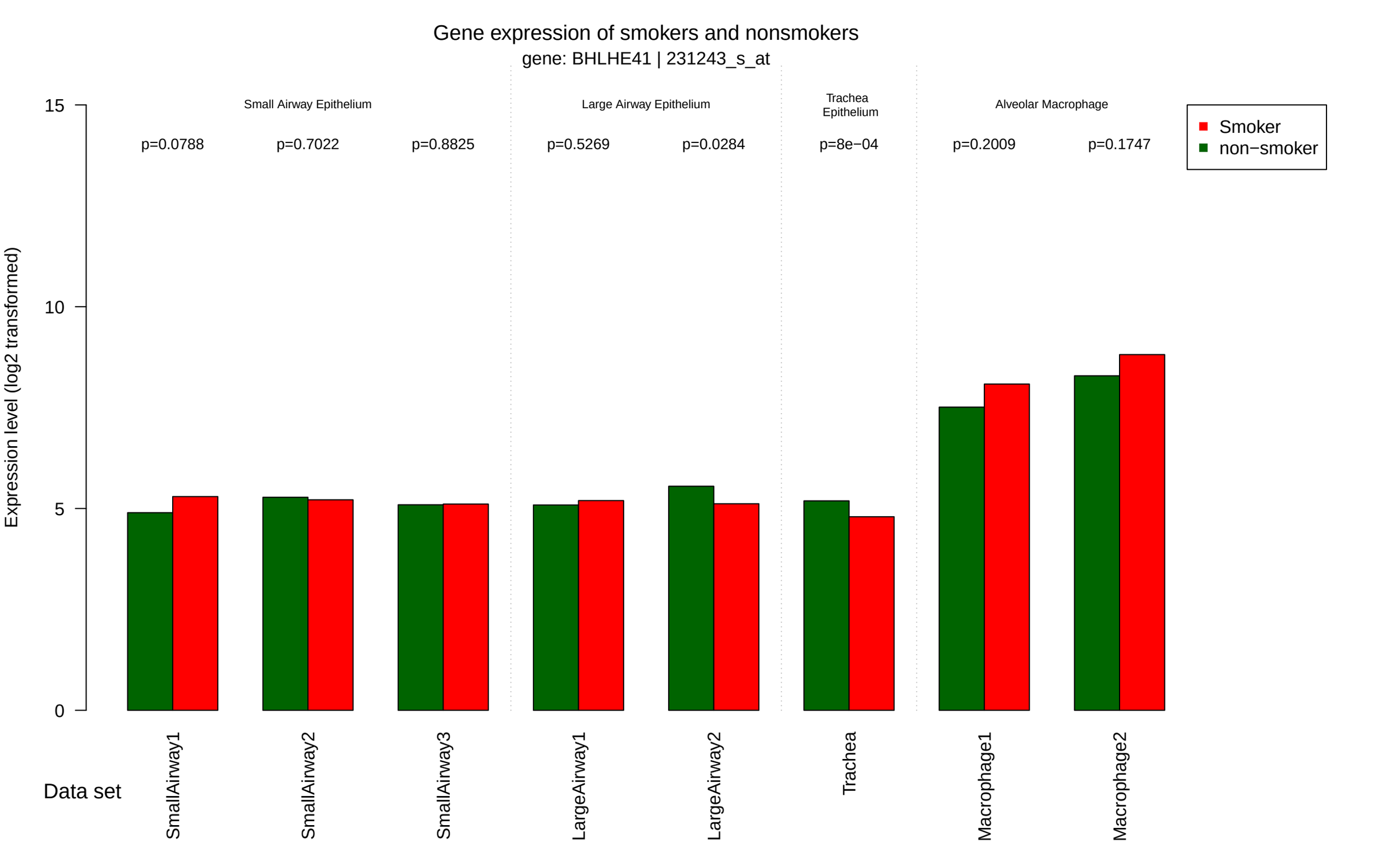
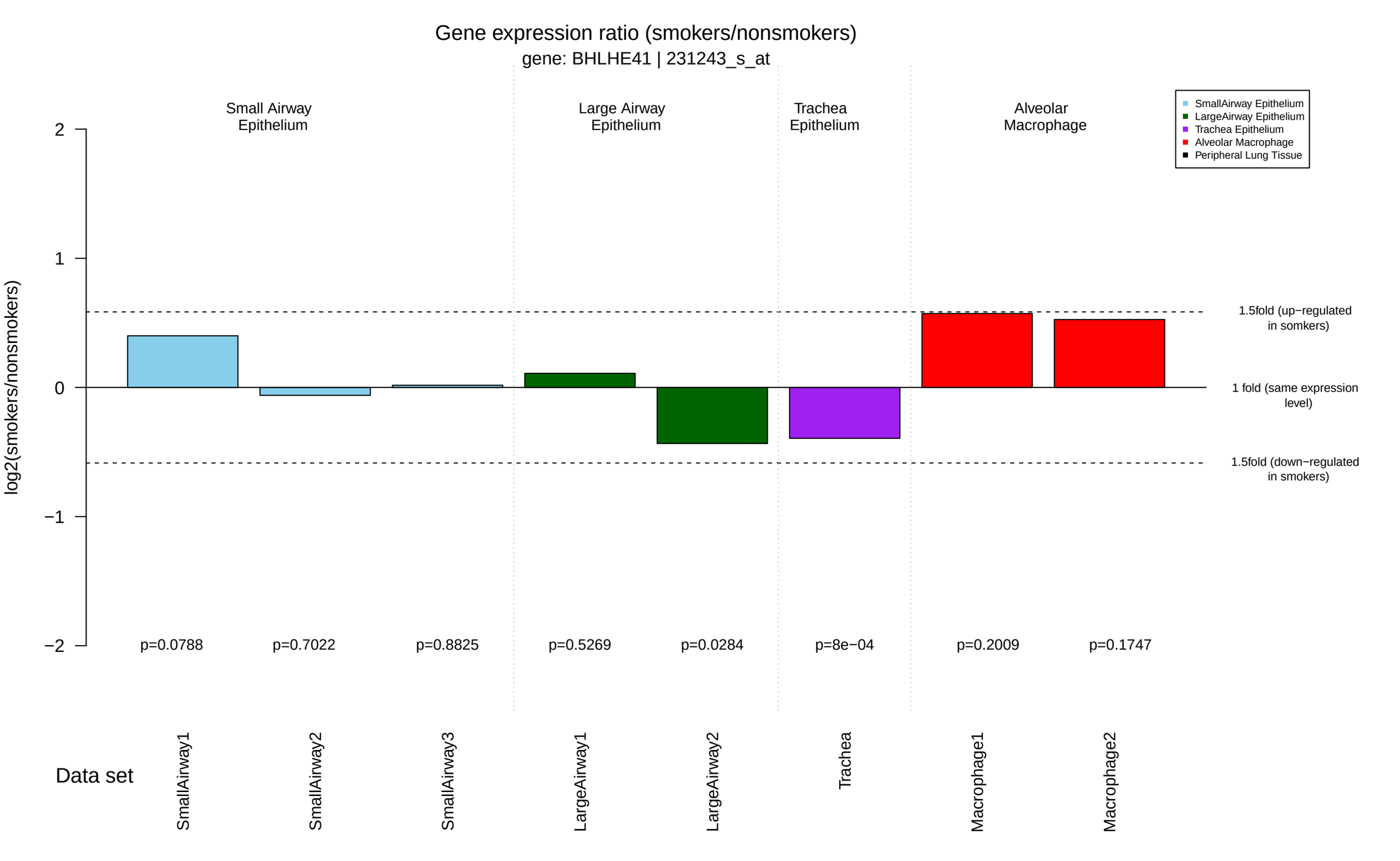
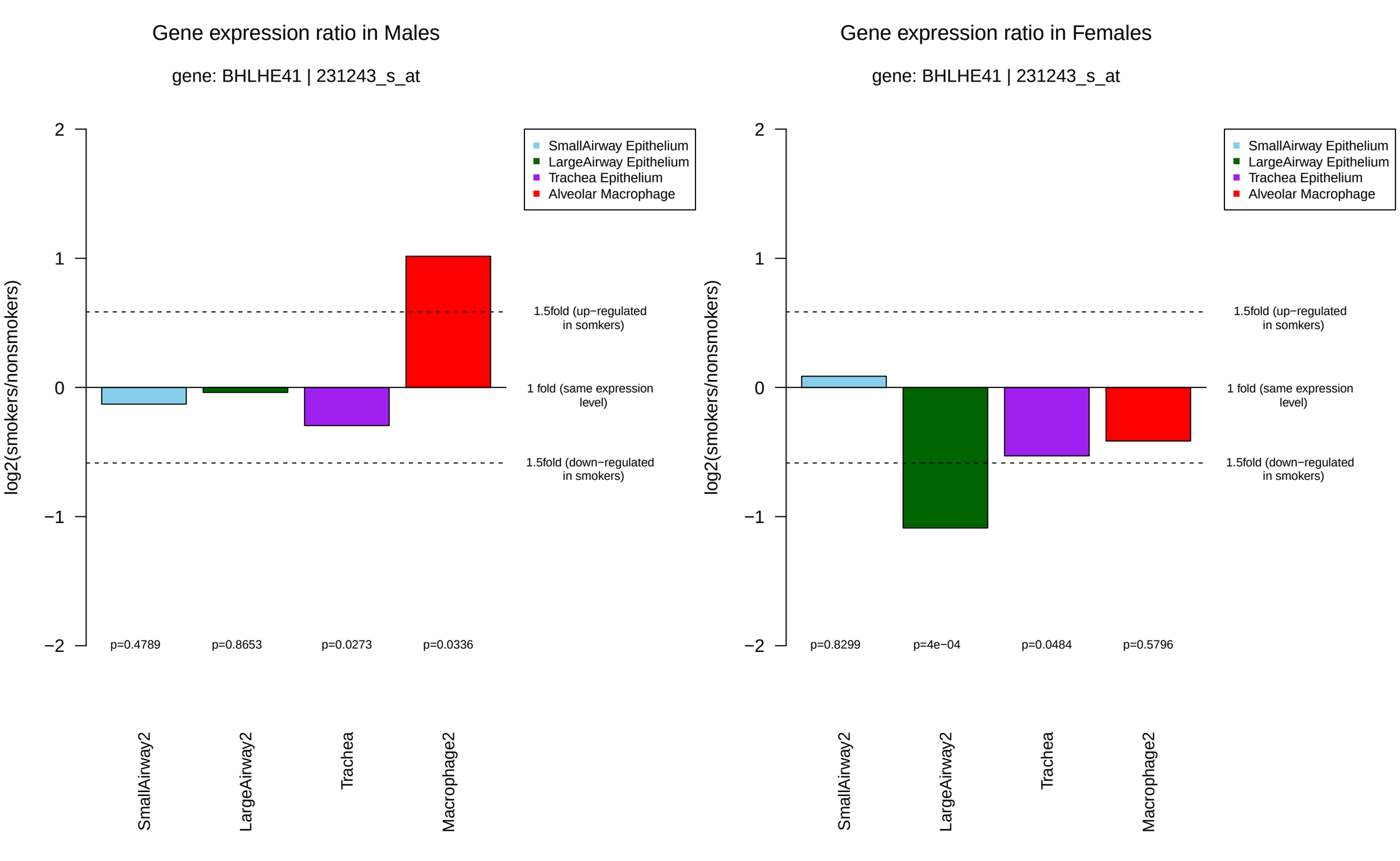
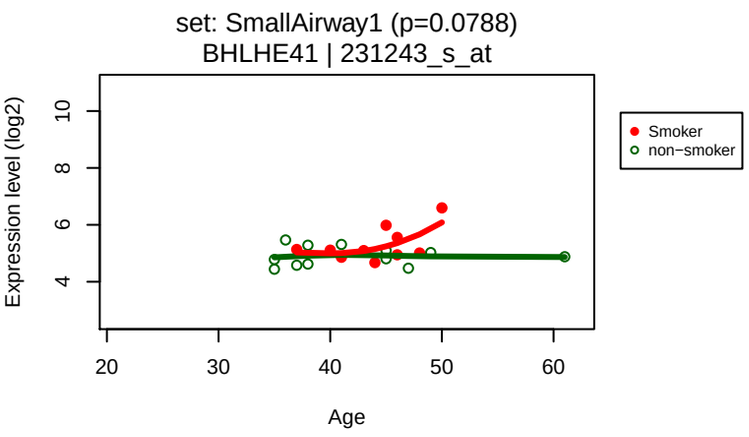
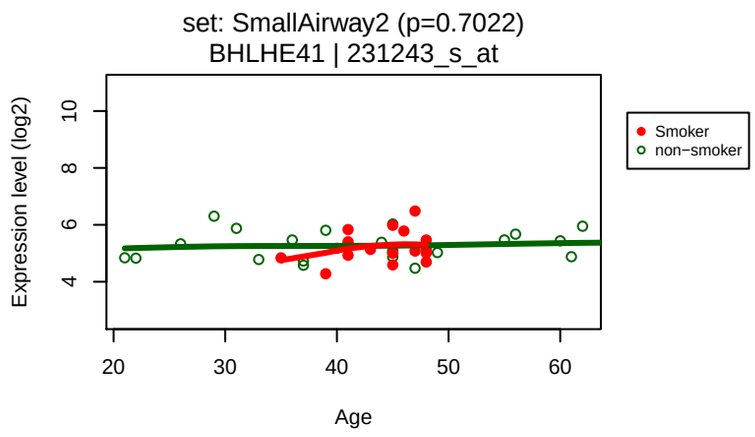
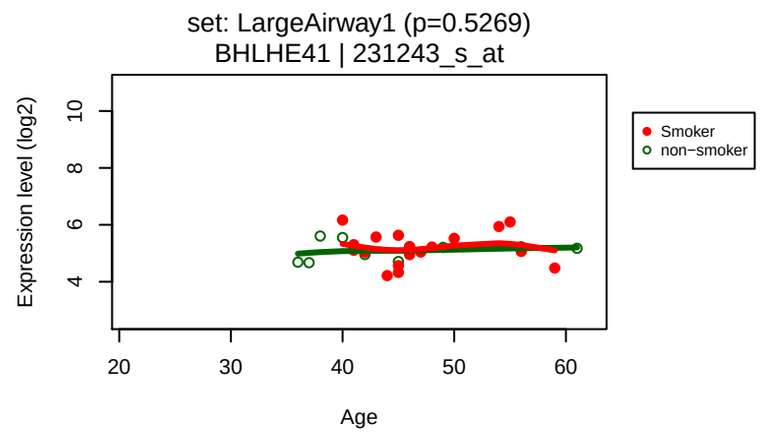

Gene expression of smokers and nonsmokers
gene: BHLHE41 | 231243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0788
p=0.7022
p=0.8825
p=0.5269
p=0.0284
p=8e−04
p=0.2009
p=0.1747
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

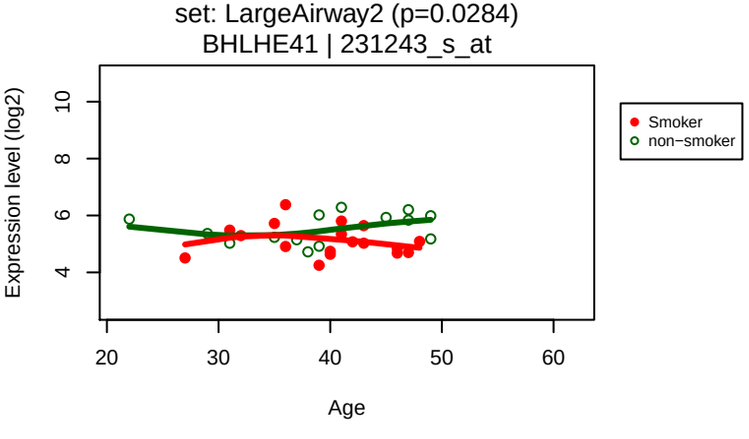

Gene expression ratio (smokers/nonsmokers)
gene: BHLHE41 | 231243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0788
p=0.7022
p=0.8825
p=0.5269
p=0.0284
p=8e−04
p=0.2009
p=0.1747
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

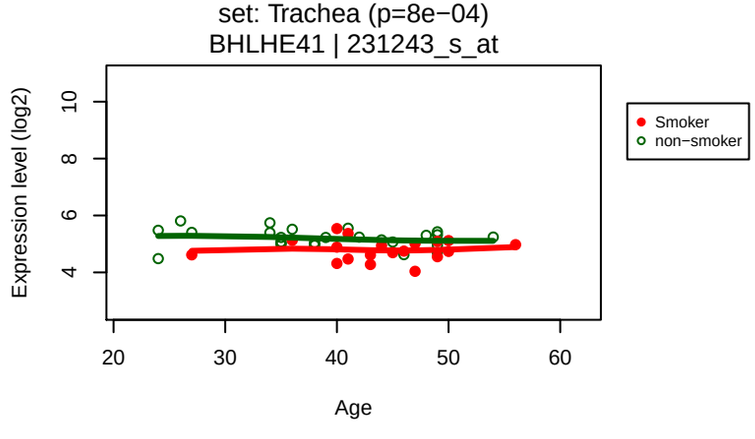

Gene expression ratio in Males
gene: BHLHE41 | 231243_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4789
p=0.8653
p=0.0273
p=0.0336
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BHLHE41 | 231243_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8299
p=4e−04
p=0.0484
p=0.5796
SmallAirway2
LargeAirway2
Trachea
Macrophage2

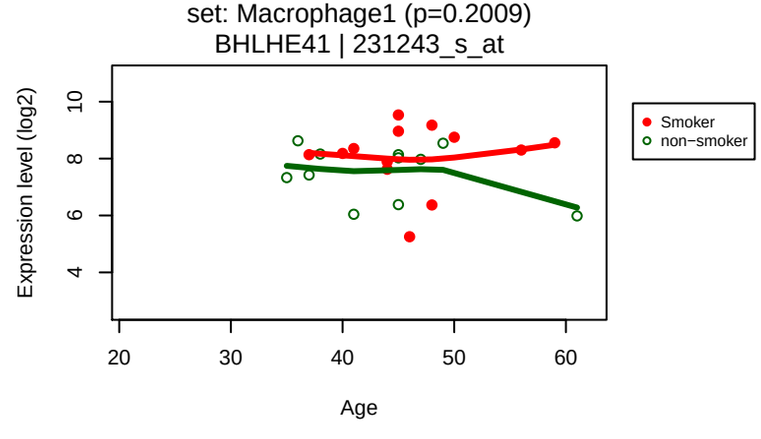

set: SmallAirway1 (p=0.0788)
BHLHE41 | 231243_s_at
Expression level (log2)
Age
Smoker
non−smoker

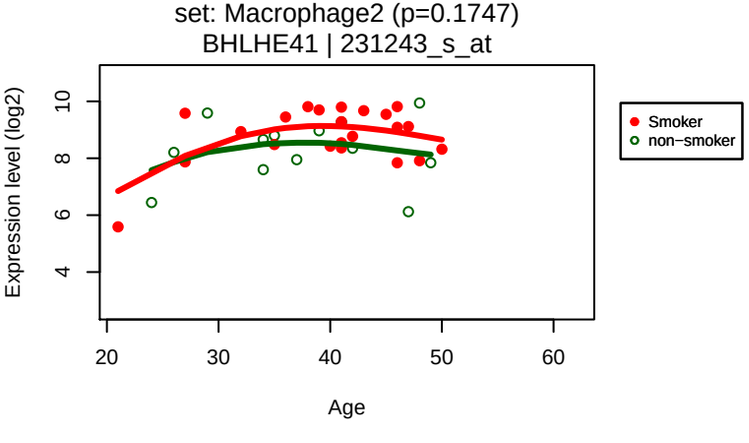

set: SmallAirway2 (p=0.7022)
BHLHE41 | 231243_s_at
Expression level (log2)
Age
Smoker
non−smoker

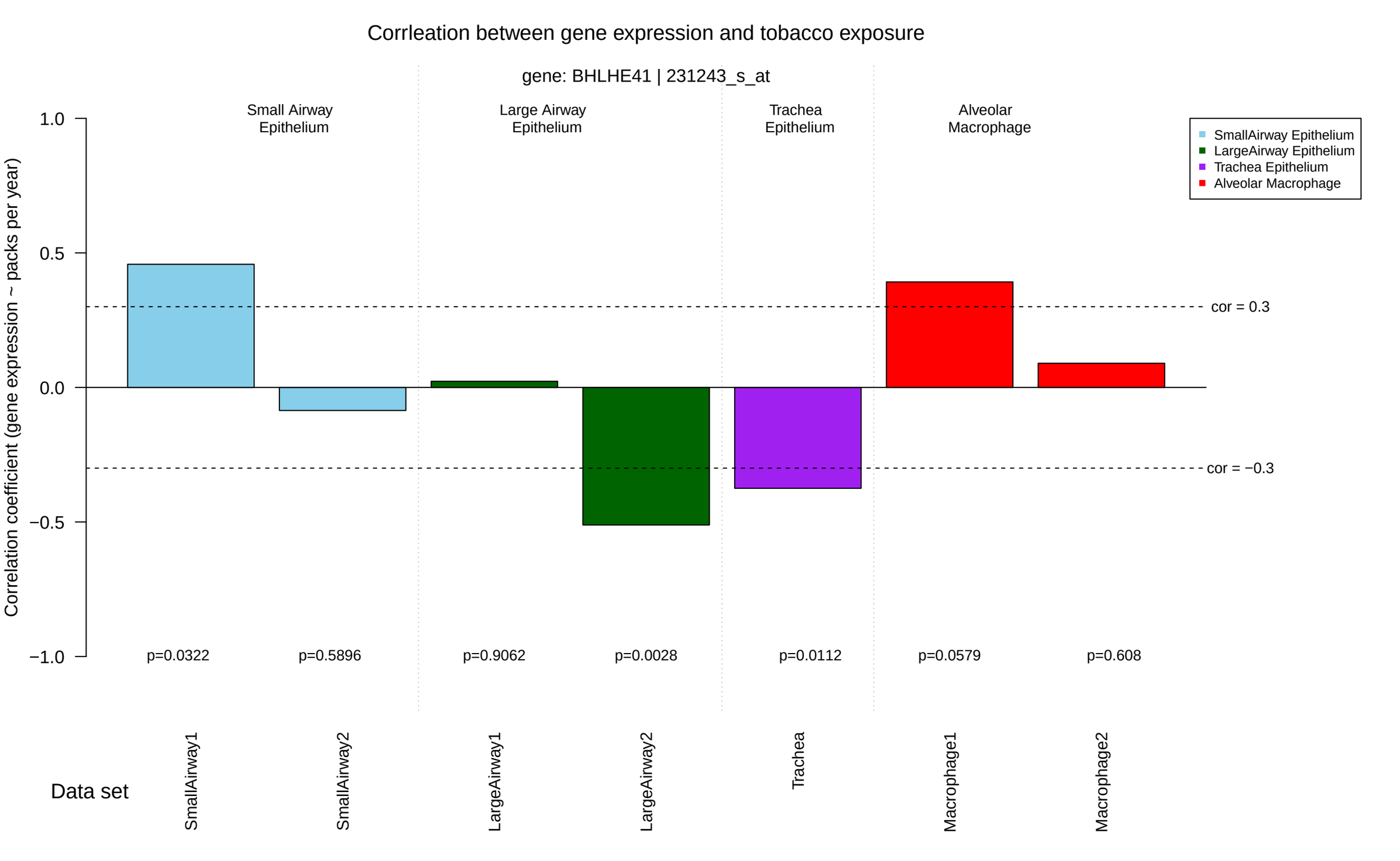

set: LargeAirway1 (p=0.5269)
BHLHE41 | 231243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0284)
BHLHE41 | 231243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=8e−04)
BHLHE41 | 231243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2009)
BHLHE41 | 231243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1747)
BHLHE41 | 231243_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BHLHE41 | 231243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0322
p=0.5896
p=0.9062
p=0.0028
p=0.0112
p=0.0579
p=0.608
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2